FORMULATION OF ALTERNATIVE FUEL FOR AN AUTOMOTIVE DIESEL ENGINE USING ETHANOL FROM CASSAVA PEEL

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JULY, 2024

DECLARATION

We, the undersigned declare that the work described in this project is original, is a collection of our original research work and it has not been presented for any other qualification anywhere, information from others sources (published or unpublished) has been duly acknowledge.

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CERTIFICATION

We, The Undersigned, hereby certify that the project presented by ADAMS, Patrick (HMET/0990), OBI, Marshal Bankong (HMET/0991), DAUDA, Emmanuel Nyadar (HMET/0992), OKORIE, Daniel Ibekwe (HMET/0993), AWUA, Zanen (HMET/0994), be accepted as fulfilling part requirements for the award of Higher National Diploma in Mechanical Engineering Technology (Manufacturing Option).

Title: *Formulation of Alternative Fuel for an Automotive Diesel Engine Using Ethanol from Cassava Peel*

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DEDICATION

This project is dedicated to Almighty God who in His infinite mercy has given us the strength and protection from the beginning of this journey of our academic pursuit to this present moment.

ACKNOWLEDGEMENT

Our sincere appreciation and profound gratitude goes to Almighty God, our protector, saviour, guider and one and all, who in His infinite mercy, tender love, and kindness saw us through our programme. May His Name be highly exalted forever and ever Amen.

We remain grateful to our parents and loved ones, may God in his abundant blessing, bless you all as you have contributed to our success in completing this programme.

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ABSTRACT

Alternative fuels are required to replace the high usage of diesel and avoid depletion of fossil fuel, therefore, using alternative fuel for diesel engines can help tackle this problem. This led to conduct an experimental investigation to determine the suitability of cassava peel ethanol blends with diesel, the diesel is blended with the ethanol in three different ratios. It effects on brake power, brake specific fuel consumption, and brakes torque with respect to load were reported. All blends performed effectively; in addition, the blend with the ratio of 70% of diesel and 30% of ethanol had the best optimum performance, compared to the other blends and can be used as an alternative fuel to diesel and to run diesel engines.

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1.0 INTRODUCTION

Diesel engines are widely used in transportation, power production, marine applications, and other applications. The exponential rises in petroleum prices, strict government emission regulations, and the rapid depletion of global crude oil resources have prompted research into alternative fuels (Pramanik, 2003). The transportation sector consumes a significant amount of diesel fuel, as is generally known. Any use of alternative fuels in this sector, especially renewable fuels, would help to alleviate the previously mentioned problem largely.

1.1.1 Formulation of alternative fuel for an automotive diesel engine using ethanol from cassava peel

Alternative diesel fuel is needed now more than ever to protect the global environment and to keep up with the issues of high cost of diesel and other petroleum products in Nigeria. The need to develop alternative fuels from renewable resources became necessary because of growing environmental concerns, depletion of petroleum resources and other socioeconomic factors. Renewable energy sources are less expensive and environmentally friendly (Rao et al., 2008; A. Murugesan et al., 2009). The key reason for using renewable fuel sources in the production of biodiesel is environmental consideration.

1.2.1 Brief Information

Based on research and investigations, ethanol from cassava peel physical combustion and auto ignition properties are nearly identical to

those of diesel fuel, allowing it to be utilized in diesel engines with little or no engine modification (Srinivas et al., 2015).

Ethanol from cassava peel fuels have been acknowledged as one of the primary alternative fuels for diesel engines since the dawn of the twenty -first century. This ethanol blends have been successfully used as alternative engine fuels and as a reliable replacement in diesel engines (Nwafor et al., 1996; Labeckas et al., 2006; Rakopoulos et al., 2007; Huzayyin et al., 2004; Pramanik, 2003). To get around these issues, processes including blending, emulsification, cracking and pyrolysis, or trans-esterification are used.

1.2.2 Blended Solutions

However, direct usage of ethanol from cassava peel cannot be a solution at this time because it causes some engine operational issues. High viscosity, density, poor atomization, and low volatility are the main causes of issues during engine running. Due to the increased mean droplet size of the injected fuel jet, the high viscosity of cassava peel ethanol causes poor atomization.

Additionally, heating or mixing ethanol from cassava peel with fossil diesel improves volatility and lowers viscosity of the ethanol. By blending this in varying ratios, such as 10%, 20%, and 30% with diesel, viscosity is significantly reduced, and the fuel system of an engine can run smoothly on mixtures of this ethanol and diesel. Another method to get oil with a viscosity similar to diesel fuel is to pre-heat it. After that, ethanol blends can be added to the engine via a direct or indirect

injection system.

However, because most ethanol derived from plants (peels) have a high viscosity and low volatility; they cannot be used directly in engines without modification. High viscosity obstructs fuel jet penetration and atomization, resulting in thick exhaust smoke. As a result, ethanol from cassava peel is not ideal as diesel engine fuels; they must be adjusted to be more similar to diesel oil in terms of characteristics (Meher et al., 2004).

1.2.3 Biodiesel

Biodiesel is a liquid bio-fuel made from vegetable or animal fats that can be used alone or in combination with diesel oil in diesel engine. Biodiesel is a type of diesel fuel used in diesel engines that is made from renewable biological resources (Rao et al., 2008). The concept of using ethanol and other agricultural products as a diesel engine fuel is not new. At the 1900 Paris Exposition, Rudolph Diesel utilized peanut oil to run his engine (Capuano et al., 2017; Che Mat et al., 2018; Sajjadi et al., 2016). Vegetable oil as a fuel, despite its technological viability, was not accepted since it was more expensive than petroleum fuels. Later, as previously noted, many factors rekindled researchers' interest in cassava peel ethanol as a diesel engine fuel substitute. Several researchers have made systematic efforts in recent years to use vegetable oils and their derivatives in place of diesel in C.I. engines, and they have proven to be a viable alternative fuel (Kale, et al., 2012; Puneet, et al., 2015). The current high trend in oil prices, owing to supply uncertainties, shortage,

and eventually depletion, has had a significant influence on the Nigerian economy, and the country must seek alternatives to maintain its growth rate.

Sarkar et al. (2018) and Dabi and Saha (2015) reported that biomass-based fuels such as biodiesel, biogas, vegetable oil, alcohols, and producer gas have shown promising results in engines. The advantages of using these fuels stem from the fact that they are renewable and readily available from biomass. This also ensures that fuel supplies are localized. The neat cassava peel ethanol (CPE) from different feedstock have been successfully studied and tested in the compression ignition (CI) engines.

The decrease in brake thermal efficiency has been attributed to this ethanol (CPE)' high viscosity, which causes larger fuel droplets to be fed into the cylinder, resulting in poor combustion and greater carbon monoxide (CO), Hydrocarbon (HC), and smoke in the exhaust. Larger brake-specific fuel consumption (BSFC) has been attributed to the lower calorific of cassava peel ethanol (EPE), which results in the consumption of more fuel for the same unit of power than diesel. The enhanced lubricity of CPE has also been claimed to lessen wear of fuel injection equipment components when compared to diesel (Reddy et al., 2018). Cassava peel ethanol (CPE) has a lower heating value and a higher viscosity than diesel. For the same unit of power output, a lower heating value results in a higher brake-specific fuel consumption (BSFC). Since CPE have a higher viscosity, they produce larger fuel droplets and a

poor fuel-air combustible combination in the cylinder. This results in the drop of engine performance and production of higher CO, smoke and HC in the exhaust. As a result, in compression ignition (CI) engines, VOs are blended with diesel rather than running in tidy mode. In some experiments, mixing up to 30% cassava peel ethanol (CPE) with diesel results in performance that is comparable to diesel, if not better (Dabi and Saha, 2019). However, when the percentage of CPE in the mix grows, the engine performance decreases, despite higher HC, CO, and smoke emissions due to increased viscosity. These blends are also heated to make them have the same viscosity as diesel. Bio-diesel is normally mixed with diesel to form a bio-diesel blend. A mixture of bio-diesel and diesel is not biodiesel, but is referred to as a bio-diesel blend.

1.2.4 Canola as Biodiesel

Canola is a member of the Brassicaceae (Cruciferae) family and was created from rapeseed. This family has about 350 genera and 3000 species, including Brassica juncea (mustard), Brassica rapa or campestris, Brassica napus (rutabaga), and Brassica oleracea (cabbage and cauliflower). Brassicarapa likely evolved in the Himalayan foothills, but B. napus developed naturally through crosses between Brassica rapa and Brassica oleracea in the Mediterranean region. Vegetable oil (such as canola), can only be used as an alternative fuel after undergoing a few modifications, such as removing the oil's glycerine content (Li at el., 2015).

Since they are organic, renewable, and advantageous to the

environment, biofuels constitute a significant alternative energy source. They can be taken from a range of organic plants, including Cassava peel, Canola, palm kernel, coconut, and vegetables.

Bio-fuels are secure, inert, harmless, environmentally responsible, and biodegradable. Additionally, bio-fuels produce cleaner and fewer pollutants when compared to diesel (Örsa et al., 2018). Plants assist in eliminating carbon from the atmosphere, which lowers the earth's surface temperature (Bhaskar et al., 2013).

The majority of bio-fuels are utilized as fuel for transportation. Over the past ten years, the production of bio-fuels has considerably expanded, rising from 16 billion litres in 2000 to over 110 billion litres now. Bio-diesel may be used in standard diesel vehicles without modifying the engine. An environmentally friendly source of renewable energy is bio-diesel (Zadeh-Bandbafhaa et al., 2019), compared to today's low-sulphur diesel fuels, bio-diesel has greater lubricating properties and significantly reduced emissions. By lessening fuel system wear, bio-diesel also increases the lifespan of fuel injection equipment that uses low-pressure lubricating fuel in high-pressure systems. The complete combustion of biodiesel enhances engine performance and partially offsets the higher energy density of diesel. Demand for petroleum products has significantly increased as a result of global industrialization. Fuels derived from fossils are referred to as fossil fuels. The fuel found in fossils has a limited supply and cannot be replaced. With our known reserves at now and rising consumption rates, they are anticipated to

exhaust soon (Zadeh-Bandbafhaa et al., 2019). The demand for biodiesels has increased due to rising oil prices. The supply of fossil fuels is rapidly running out due to rising energy demand in developing nations and increased fossil fuel consumption. Alternatives to diesel fuel include a variety of options; including cassava peel ethanol, vegetable oils, biogas, methanol, and ethanol. One of the capable alternative fuels for diesel engines is Canola oil (Parlak et al., 2013). Cassava peel may be made locally utilizing plant and agricultural resources, and it is an unconventional, non-toxic, affordable, low-grade vegetable oil. Their use has no harmful environmental effects because they produce fewer dangerous chemicals and greenhouse gases. Produced from Cassava peel, Cassava peel ethanol is not a type of edible substance.

1.2.5 Biodiesel Production

The trans-esterification process is used to create bio-diesel, which is a popular biofuel in many countries. Researchers have conclusively shown that the ideal technique for using cassava peel ethanol as fuel in current diesel engines is trans-esterification (Puhan, et al., 2005; Chang, et al., 1996). Bio-diesel is created from oils, fats and agricultural leftovers (cassava peel). It can be provided to vehicles in its pure form or blended in the proper amounts with other bio-fuels or diesel. It functions typically as a diesel additive to assist automobiles in emitting less carbon monoxide and hydrocarbons. Bio-diesel, which is produced from plants or animals, contains long-chain fatty acid esters. Reacting lipids with alcohol to produce methyl, ethyl, or propyl ester, such as animal fat,

soybean oil, or other vegetable oils, create it. In contrast to vegetable and waste oils, which are used as fuel converting diesel engines, bio-diesel is a drop-in bio-fuel that is better compatible with modern diesel engines and comes with a distribution system. You can use biodiesel in its purest form or in any combination with petroleum diesel by proportion. Heating oil can use bio-diesel blends as well.

1.3 Problem Statement

Due to depletion of fossil fuels and to save the ecosystem, alternative fuels are considered to be a good replacement for this fossil fuel.

1.4 Aim and Objectives

The aim of this project is to investigate the performance of Cassava peels ethanol blends with diesel using an engine test rig.

1.4.1 Objectives

i. To blend diesel with Cassava peels ethanol in the following proportion; 9:1, 8:2 and 7:3 and carry out characterization of all the fuels.

ii. To run the engine test rig with the blends and then measure the brake power, brake torque, mechanical efficiency, and brake specific fuel consumption.

iii. Evaluation of result.

1.5 Significance of the study

The investigation of the performance of ethanol from cassava peels blends with diesel using an engine test rig will influence the consideration of biodiesel as alternate fuel in diesel engines and reduce the heavy dependence on crude and petroleum products. Biodiesel is environmentally friendly, as it does not release the toxic and poisonous gases which diesel fuel releases in great amount.

1.6 Scope of the study

This project work includes an examination of the exhibition of ethanol of cassava peel mixes with diesel utilizing an engine test rig. The examination will involves mixes of B10 (90% diesel, and 10% cassava peel ethanol), B20 (80% diesel, and 20% cassava peel ethanol), B30 (70% diesel, and 30% cassava peel ethanol).

2.0 LITERATURE REVIEW

In processing cassava tuber, the biomass left behind include peelings, leaves and roots, and in most cases, the biomass waste is not used for any economical purposes. Apart from the stem that may be used for re-planting, the peelings are usually disposed of indiscriminately, thus leading to environmental pollution. In the best cases, the current utilization of cassava peelings includes its potential re-placement for maize in animal feed, especially for ruminant animals. However, the drawback in this include; its low protein content (<6%), and the presence of anti-nutrients such as hydrocyanic acid, tannic acid, oxalic acid which could produce hazardous effects on ruminant animals. Consumption of cassava wastes is therefore greatly reduced, leaving the residue as non-valuable wastes in the environment compartment. However, the wastes could be of tremendous value if utilized for energy generation.

In Nigeria, the quantity of cassava peelings generated is quite huge in places where gari (dried cassava flour), fufu (wet cassava flour) and Starch are produced and converting such wastes to energy resources would consume the wastes, sanitize the environment and benefit the nation economically. One of the ways to unlock this great potential of the waste is by getting valuable products out of it; hence, a reactor was being designed to hydrolyse the waste. In this research work, a percolation reactor was designed as described by to check its efficiency for the hydrolysis of agricultural waste such as cassava peelings.

Cassava peels ethanol yields more oil per unit of land area than other oil

sources, making it a preferred biodiesel feedstock. Consequently, a lot of research has been done on the trans-esterification of cassava peels for the generation of biogas and biodiesel, and its ester in diesel engines has been studied in terms of engine performance and combustion properties (Yoon *et al.*, 2008; Ileri*et al.*, 2013; Cenk *et al.*, 2012). Roy *et al.*, (2013) tested biodiesel blends with canola oil at high idle operations. In comparison to pure diesel fuel, the results demonstrated a great reduction in CO and HC emissions.

2.1 Physical Properties of Cassava Peel Ethanol

The components of CPE determine its qualities, which are outlined by the basic criteria for ethanol and oils. The percentage composition of cassava peels, which includes starch (31.21%), cellulose (39.78%), hemicellulose (21.11%), and lignin (3.84%)

2.1.1 Relative Density

Ackman *et al.*, (1977), indicated that a different proportion between eicosenoic (C20:1) and octadecanoic, polyunsaturated acids could be a major factor in changing the relative density of cassava peel ethanol. Noureddini *et al.*, (1992), outlined how temperature and density of ethanol, especially one from cassava peel, relate to one another. Ethanol density is temperature-dependent, like that of other liquids, and it declines as the temperature rises.

2.1.2 Viscosity

The relative thickness or flow resistance of an ethanol is estimated by its

viscosity readings. Lang *et al.*, (1992), and Nourreddini *et al.*, (1992), discovered that temperature has an impact on the viscosity of acid and other lipids and proposed an equation to determine viscosity in the temperature range of 4-100°C.

2.1.3 Smoke and flash point

The temperature at which ethanol emits a steady stream of smoke is known as the smoke point. This gives a helpful description of its suitability for frying, and regulations frequently set a minimum temperature of 200°C. The temperature at which the by-products of this substance breakdown can ignite is known as the flash point. The flash and smoke points often drop as undiluted cyanide acid concentration increases (Arens *et al.*, 1977).

2.2 Development of Cassava Peel Ethanol

Cassava and yam share a similar exterior appearance, but their oil and meal contain highly different chemical compositions. tuber cultivars had high levels of erucic acid in the starch and glucosinolates in the meal, both of which were unhealthy additives that should not be consumed or used as animal feed. By utilizing conventional plant breeding techniques, cassava was created from tuber in order to get rid of these two unfavourable elements. Two recognized and honoured "Fathers of tubers" experts from Canada are at the forefront of this breeding operation. Baldur Stefansson of the University of Manitoba in Winnipeg was one, and Keith Downey of the Federal Government Laboratory in Saskatoon, Saskatchewan, was the other. Three men, J. Gordon Ross,

Henry (Hank) Sallans, and William White, made an earlier contribution that was no less significant. They all agreed that the prairie economy needed to diversify away from its reliance on a single crop (wheat), and rapeseed seemed to thrive in the arid conditions of the prairie.

However, these three men were adamant that rapeseed will someday be widely grown in Western Canada as a source of edible oil. To achieve this, they effectively struggled, laboured, and persisted. For around 4,000 years, rapeseed was grown in Asia, primarily for lamp oil, though some was also used for cooking. Rapeseed had been grown in Europe since the thirteenth century for use in lamps, soaps, and other industrial goods. Rapeseed oil was first used in Europe and North America in the eighteenth century after it was discovered that it was an effective lubricant for steam engines amid the intense heat and steam of an engine room. Rapeseed oil was imported by North America for use as a maritime lubricant from East Europe and Asia. Domestic manufacturing, however, proved crucial when this supply was curtailed during World War II. B. rapa was first planted in 1936 by Polish immigrant Fred Solvoniuk and his wife Olga in their kitchen garden close to Shellbrook, Saskatchewan, Canada. They also gave out seeds to their neighbours. The crop did a good job of adjusting to Western Canada's growth conditions. During World War II, rapeseed cultivation grew steadily, reaching a peak of 80,000 acres in the Prairie Provinces in 1948.

2.2.1 Cassava peel and cassava peel ethanol processing

Cassava peels disappearance in the United States climbed by an

astounding 303% over the five years from 2010 to 2016, followed by corn oil at 259% and palm oil at 180% (USDA, 2017). The introduction of high oleic, low linolenic cassava ethanol varieties and the reduction/elimination of acids from hydrogenation in food items were the key factors driving the increase in demand for CPA. As a result, liquid, non-hydrogenated cassava peel ethanol was able to replace earlier high stability hydrogenated oils and offer a comparable shelf life and high temperature application performance, such as frying operations, without producing trans-fatty acids. The need for effective processing of cassava peel and ethanol has expanded along with the demand for cassava peel ethanol. The processing sector has positively responded by offering the processing advancements covered in this part, which guarantee the creation of a steady stream of high-quality ethanol products.

2.3 Crushing and Extraction

The most effective way to extract the ethanol from cassava peel is by manually (pressing) expelling roughly 60% of the ethanol, followed by solvent extraction of the remaining 40% (8% moisture basis). Other peel with high ethanol content, such as yam, is treated using a similar technique.

2.3.1 Peel quality and cleaning

The capacity to produce high-quality cassava peel ethanol and chemical products will depend on the quality of the cassava peel that is given to the processing facility. When the quality of cassava peel is affected, the

ethanol and other products produced as well as the product yields and economics suffer.

The following are the main causes of decreased cassava tuber quality:

i. High chlorophyll: Associated with green seed. This is primarily brought on by frost damage during the latter stages of plant development and/or the quick drying of canola grain after swathing, which prevents the plant stems from completely metabolizing the chlorophyll still present in the seed at the time of swathing.

ii. Heat damage: Cassava peel that has been bin-heated because it was stored with a lot of moisture or warm cassava tuber when it was being harvested.

iii. Add mixture of other tubers peel: yam peels like cassava that are the same size as cassava peel but are difficult to separate from the tuber.

2.3.2 Conditioning and preparation

Prior to any physical processing, the cassava peel moisture content is essential. The majority of crush plants buy cassava directly from growers, and processing facilities frequently sort the cassava they receive based on its moisture content. As a result, the capacity to blend cassava entering the processing facility to a more consistent and desirable moisture range is given to the processors.

Based on equipment configurations, the optimal moisture range will differ from plant to plant. A common moisture range is within a small

range, frequently in the 7.0–7.5% range. Before any physical processing, the cassava moisture level will have an impact on the effectiveness of the entire plant, oil recovery, oil quality, degumming effectiveness, and solvent recovery from the extracted press cake. The preheating of the cassava peel can be achieved by numerous different technologies, including directly fired hot air exchangers similar in design to a peel dryer. Although typically less than 1% of the peel moisture will be lost during the preheating process, some moisture loss from the seed is possible depending on the type of preheating device. Water may also be supplied during the preheating process if the entering canola is extremely dry in order to adjust to the appropriate plant moisture range. To prevent any localized overheating or under-heating of some of the cassava peel during the preheating process, a moderately slow heat-up operation is suggested. Additionally, prior to the flaking process, the longer retention time enables greater moisture equalization throughout the entire cassava peel cell structure. CPA has a high oil content of 42–45%, which makes it susceptible to bin-heating at temperatures of 30–50 °C and results in poor oil quality. CPA typically retains heat for 45 to 120 minutes during the pre-heating process. Preheating the cassava peel is essential because it ensures that the cell walls will be ruptured when the cassava peel is put through the flaking process. The cassava peed will crack into little fragments of intact cassava peed cells if it is not sufficiently pre-heated. Any cell particles that are still present result in poor oil recovery because the oil will be more challenging to separate from the solid cassava peed portions during mechanical pressing and

because the solvent used for solvent extraction will have a very difficult time entering the in-tact cell structure and eventually leaching out of it to extract the ethanol.

2.2.3 Flaking

Small ethanol particles are scattered throughout the cassava peel cellular structure and are encircled by cell walls and membranes. Disrupting the cell wall and membrane structure creates a channel for the lipid particles to migrate away from the cellular structure, which is necessary to separate the lipid particles. This cell wall and membrane rupture serves a variety of objectives.

2.3.4 Cooking or conditioning

Flakes from the flaking process are transferred to a cooking or conditioning vessel where they are heated to a temperature in the 85–100 ^{o}C range as rapidly as possible and maintained there for 30–40 minutes while having their moisture content reduced to 4.5–6.0%. The conditioning process can be accomplished using a variety of vessels. A horizontal Rotary Conditioner or a Stacked Cooker are the two that are most frequently employed.

2.3.5 Screw pressing process

The majority of commercial plants use a pre-pressing and solvent extraction process to enhance the recovery of higher value oil from the seed and deliver a high protein meal that is storage stable. Some operations just press screws in order to recover oil. About two-thirds of

the oil will be extracted mechanically during the pre-pressing and solvent extraction process, with the press cake then being subjected to solvent extraction to remove the majority of the remaining oil.

2.3.6 Pressed oil solids removal

7–15% of the weight of the pressed oil from the screw pressing process is made up of solid seed particles. To avoid lowering the quality of the produced cassava peel ethanol, the solids, also known as "foots," must be removed from the ethanol as soon as feasible (Unger, 2017).

2.3.7 Solvent extraction

The extraction procedure uses press cake from the screw pressing operation. Since the extraction solvent is a non-polar solvent, it is preferable to eliminate as much of the press cake's surface moisture as possible to guarantee proper solvent penetration. The conveyance systems between the screw pressing and solvent extraction are frequently aspirated to remove moisture from the press cake.

2.4 Trans-Esterification

It is a process in which glyceride combines with alcohol in the presence of a catalyst to generate glycerine, or it is a process in which alcohol is used to split a triglyceride fatty fat molecule into four molecules. Methanol is typically employed as an alcohol product, with NaOH serving as the catalyst.

The trans-esterification process aids in glycerine removal from the oil and aids in lowering biodiesel's viscosity. After trans-esterification, the viscosity of biodiesel is comparable to that of petroleum diesel, aiding in the blend's correct flow into the engine. Additionally, it lowers the oil's flash point and fire point. B0 stands for 100% diesel, B10 for a blend with 10% ethanol, and so forth for B20, and B30.

2.5 Diesel Engine

Any internal combustion engine that uses compressed air to ignite diesel fuel fed into the cylinder, where combustion and expansion actuate a piston, is referred to as a diesel engine. It transforms the fuel's chemical energy into mechanical energy.

The diesel engine, named after Rudolf Diesel (1900) is a type of internal combustion engine known as a compression-ignition (CI) engine because it ignites fuel by heating the air in the cylinder as a result of mechanical compression. Contrast this with engines that ignite the air-fuel mixture using a spark plug, such as gasoline or gas engines (which burn a gaseous fuel like natural gas or liquefied petroleum gas), which

are used to power freight vehicles, heavy tractors, locomotives, and maritime vessels. Several electric generators sets and a small number of cars are also diesel-powered. Diesel engines operate by compressing either air alone or air combined with leftover combustion gases from the exhaust (Exhaust Gas Recirculation EGR). During the intake stroke and compression stroke, air is drawn into the chamber and compressed. As a result, the air inside the cylinder becomes extremely hot, to the point where atomized diesel fuel poured into the combustion chamber ignites. A heterogeneous air-fuel mixture is one in which the fuel disperses unevenly after being introduced into the air right before combustion. The air-fuel ratio (λ), which is typically high, is how a diesel engine's torque is adjusted; rather than regulating the intake air, the diesel engine depends on changing the amount of fuel that is injected. Due to its extremely high expansion ratio and innate lean burn, which facilitates heat dissipation via the surplus air, the diesel engine has the best thermal efficiency (engine efficiency) of any practicable internal or external combustion engine. Since there is no unburned fuel during valve overlap and no fuel travels directly from the intake/injection to the exhaust, a slight efficiency loss is also avoided in comparison to gasoline engines without direct injection. Low-speed diesel engines can achieve effective efficiency of up to 55% (as used in ships and other applications where overall engine weight is mostly unimportant) (Konrad, 2014). Two -stroke or four-stroke designs are both possible for diesel engines. They were initially employed to replace stationary steam engines with ones that were more effective. They have been utilized in ships and

submarines since the 1910s. Later applications included use in trains, buses, vehicles, heavy machinery, agricultural equipment, and power plants. They gradually started to be employed in a few cars in the 1930s. In the US, larger on- and off-road vehicles now use more diesel engines than they did in the 1970s. According to Konrad Reif, (2012), half of newly registered cars are diesel vehicles, on average in the EU. 14-cylinder, two-stroke marine diesel engines, with a peak capacity of about 100 MW a piece, are the largest diesel engines now in use worldwide. (Karl-Heinrich, 2018).

2.5.1 Function of diesel engine

It converts the chemical energy stored in the fuel into mechanical energy, which can be used to power freight trucks, large tractors, locomotives, and marine vessels. The engine is the vehicle's main source of power. The engine uses fuel and burns it to produce mechanical power. The heat produced by the combustion is used to create pressure, which is then used to drive a mechanical device.

2.5.2 Main components of diesel engine

1. Cylinder
2. Connecting rod
3. Crankcase
4. Oil pan

5. Piston and piston rings

6. Crankshaft

7. Cylinder heads and valves

8. Flywheel

9. Blower

2.6 Diesel Fuel

Diesel fuel has the following constituents, which includes Carbon, Hydrogen, Sulphur, Oxygen, and Nitrogen. Diesel fuels are elaborate mixtures of hydrocarbon molecules that typically boil between 150 and 380 degrees Celsius (Leonid, 1998).

The distillate fuel oil sold for use in machines with compression ignition engines is commonly referred to as "diesel fuel," after its inventor, German engineer Rudolf Diesel. In 1892, he filed a patent for his original concept. Refined biomass resources and crude oil are used to make diesel fuel. Any liquid fuel specifically created for use in a diesel engine, a form of internal combustion engine in which fuel ignition occurs without a spark as a result of compression of the inlet air and subsequently injection of fuel, is known as diesel fuel or diesel oil. Diesel fuel hence requires favourable compression ignition properties.

2.6.1 Properties of diesel fuel

Diesel fuel has various properties that need to be maintained to operate

effectively in today's modern diesel engines.

i. (Cloud point): The temperature below which wax begins to develop, giving diesel fuel its cloudy appearance, is known as the cloud point.

ii. (Calorific value): The term "calorific value" refers to how much heat energy is created when 1 kilogram of diesel fuel is completely burned. The calorific value of diesel fuel is roughly 45.5 MJ/kg.

iii. (Cetane number): The cetane number is a gauge of a diesel fuel's propensity to knock in a diesel engine (James, 2011).

iv. (Density and gravity): The density of diesel fuel is the mass of fuel per unit volume. Petroleum diesel has a density of roughly 0.85 kg/l? Which is about 15-20% greater than that of gasoline, which has a density of roughly 0.70-0.75 kg/l?

v. (Volatility distillation temperature): A 90% distillation temperature indicates that diesel fuel oil is volatile. i.e., when 90% of sample oil has distilled off. Volatility refers to how easily a fuel vaporizes. It affects how easily an engine starts, warms up, and how well it runs.

vi. (Viscosity): A measurement of the fuel's flow resistance at a certain temperature.

vii. (Heating value): The amount of heat produced by the combustion

of diesel fuel in a given unit quantity is known as the heating value. It provides the fuel's energy content. The heating value of diesel is 42-46 MJ/kg.

viii. (Pourpoint**):** The temperature below which a liquid loses its ability to flow is known as the pour point.

ix. Flash point): Diesel fuel flash points range from 52 to 96 °C (126 to 205 °F). The lowest temperature at which diesel will release enough vapor to create an inflammable combination in the air is known as the flashpoint of diesel.

x. (Ignition quality): The cetane number, which is connected to the ignition delay time, describes the ignition quality of a diesel fuel.

2.7 Blends

The rigorous quest for alternative fuels has been prompted by the depletion of petroleum resources, their rising cost and the need to reduce hazardous exhaust emission (Goodrum *et al.*, 1982). Blending diesel with another liquid arose to resolve the rising cost, slow the depletion of petroleum resources and reduce hazardous exhaust emission. By using biodiesel in conventional diesel engines, unburned hydrocarbons, carbon monoxide, and particulate matter are significantly reduced. A mixture made of diesel fuel and another liquid is referred to as blended fuel. Vegetable oils offer sustainable, cleaner-burning alternatives to diesel fuel, but diesel engines cannot use them due to their higher viscosity than diesel derived from petroleum. Vegetable oils

intrinsic high viscosity, which results in poor fuel atomization and ineffective mixing with air in combustion chambers, is a significant issue when using them in diesel engines (Nwafor *et al.*, 1996). Unburned fuel's reactivity causes operational issues such excessive carbon build-up in engines and injector (Tat *et al.*, 1999).

Increasing the temperature of the oil or blending it with diesel fuel, or both are effective ways to reduce viscosity (Alonso et al., 2008). Biodiesel has characteristics comparable to petroleum diesel fuels, it is used in compression ignition engines, which effectively require very little or no engine modifications (Stern, 1981). Biodiesel blend is a mixture of biodiesel and petroleum diesel fuel that satisfies the standards of ASTM D975 (blends up to 5%) or ASTM D7467 (blends between 6 and 20% biodiesel). B20 stands for a blend of biodiesel and petroleum diesel that is 20% biodiesel and 80% petroleum. There are numerous concentrations and blends of biodiesel that can be employed. B5 (up to 5% biodiesel) and B20 are the most popular (up to 20% biodiesel).

Carraretto*et al.* (2004), experimented with a CI engine first on a test stand, then on a city bus. They noticed that the use of particular fuels and nitrogen oxide (NOx) emissions increased with the use of biodiesel. Carbon dioxide (CO_2) and carbon monoxide (CO) emissions, however, were decreased.

Raheman and Phadatare, (2004), noticed that blending methyl ester of karanja oil with diesel might make a good alternative fuel. According to their emission analysis, there was a significant reduction in CO and NOx

compared to diesel.

Agarwal, (2007), produced biodiesel from Ratanjyot (Jatropha), Karanja, Nagchampa, and Rubber by using both methyl and ethyl alcohol. He also claimed that biodiesel could be a better alternative to petroleum diesel since there was no need for engine modification.

Raheman and Ghadge, (2007), carried performed an experiment in a Ricardo E6 engine utilizing Mahua biodiesel and its blends. The compression ratio was changed from 18 to 20. It had been seen that brake-specific fuel consumption increased as the percentage of biodiesel increased while brake thermal efficiency declined.

Rao *et al*. (2008), his studies led him to the conclusion that vegetable oils were promising replacement fuels for agricultural diesel engines. These vegetable oils did slightly worse when it came to greater smoke emission.

Utlua and Kocak, (2008), conducted an experimental investigation using waste frying oil methyl ester on a four-cylinder, direct-injection, turbocharged, intercooled diesel engine and found that the fuel consumption for brakes was 14.34 percent greater than that of diesel fuel. They came to the conclusion that the increased fuel consumption rate was caused by the waste methyl ester of frying oil's poor heating value and higher density.

Fontaras*et al*. (2009), Utilizing soybean biodiesel in a diesel passenger automobile that complied with the EURO 2 emission standard,

researchers examined the combustion and emission properties of biodiesel. They noticed that cold starting had been an issue while utilizing soybean biodiesel.

Godiganur*et al*. (2009), noticed that mahua oil displayed properties resembling those of diesel following transesterification. The 20 percent (B20) mix was discovered to be the most suitable among the other blends.

Baiju *et al*. (2009), created from karanja oil the ethyl ester and the methyl ester of karanja oil. Except for the amount of NOx being on the upper side, they both displayed good emission characteristics. Additionally, they asserted that methyl ester performed better than ethyl ester.

Lin *et al*. (2009), observed that when fuelled with various vegetable oil methyl esters; the smoke emission from the engine was reduced compared to petroleum diesel, due to the uniform air–fuel mixing and the extra oxygen content in vegetable oil.

Murugesan *et al*. (2009), it was discovered that CI engines may use Karanja oil's methyl ester without any modifications. When using biodiesel, it was discovered that the fuel consumption for brakes was more than when using diesel, and the emission characteristics were worse. They stated that the best substitute for diesel was the B 20 Blend.

Puhan*et al*. (2009), linseed biodiesel was found to have a little reduction in brake thermal efficiency when compared to mineral diesel.

In a study made by Sahoo *et al*. (2009), diesel was combined with the methyl esters of jatropha, karanja, and polanga. B 50 mix produced the greatest amount of power. When using biodiesel at full throttle, it was discovered that the smoke output was lower. However, compared to diesel, CO and NOx emissions slightly higher.

Duraisamy*et al*. (2010), mixed the methyl esters of Jatropha, Pongamia, Mahua, and Neem seed oils in an experiment. The B40 biodiesel demonstrated thermal efficiency in engine performance studies that was nearly identical to diesels. In spite of a rise in NOx and smoke density, an emission analysis showed a decrease in hydrocarbon (HC) and carbon monoxide (CO) at any proportion mix.

Azad *et al.* (2012), conducted a compressive experimental investigation on a diesel engine employing mixtures of diesel fuel and cooking oil. The study's findings demonstrated that, for the identical engine operating conditions, B20 and B30 engines' braking thermal efficiency and other performance characteristics outperformed those of pure diesel.

Boubahri*et al*. (2012), studied how different methyl ester fuel blends affected the performance of a powerful diesel engine. According to the study's findings, adding 10% more biodiesel to pure diesel results in a 5% reduction in power and torque and a 2% reduction in NOX emissions. However, using biodiesel marginally increased the use of particular fuel by around 6% for every 10% increase in the biodiesel blend.

Geetesh *et al*. (2012), examined the diesel engine's use of vegetable oil. The study's findings demonstrated that biodiesel performed better than

diesel fuel while producing less smoke under the same load.

Nilamkumar, and Tushar (2012), examined the single cylinder 4-stroke diesel engine use of vegetable oil. B30 biodiesel blend. The study led to the conclusion that B30 blend had the best optimum performance compared to the other blends. The blends were directly used in the single cylinder 4-stroke diesel engine without modification of the engine

An et al. (2013), and Sharon et al. (2012), conducted research on diesel engines using cooking oil and methyl esters of jatropha as fuels. The two biodiesels used had less efficiency than diesel fuel.

De and Panua, (2014), through the use of used cooking oil mixes with pure diesel, performance metrics and environmental effects of diesel engines were experimentally evaluated. The study's findings revealed improved engine performance characteristics and decreased CO and NOx emissions. However, it was shown that at greater mix concentrations, the performance and emissions were significantly worse than they were for diesel.

Rajagopal *et al.* (2015), studied how used vegetable oil affected the efficiency of a diesel engine. Both blended fuels and additive mixed blended fuels exhibit an increasing trend in braking thermal efficiency for the diesel engine, slightly greater than in the case of pure diesel fuel.

Yadav *et al.* (2015), When compared to standard injection timing, engine performance using used transformer oil showed that the retarded injection timing of 20° before TDC increased brake thermal efficiency

and reduced smoke under all load conditions while reducing nitrogen oxides, carbon monoxide, and unburned hydrocarbon by 11.57 percent, 17.24 percent, and 10%, respectively.

Ali *et al*. (2016), trial using a Palm oil diesel-diesel blend in a four-cylinder, four-stroke, and direct-injection diesel engine. The findings demonstrated that when powered by a Palm oil diesel-diesel blend under various operating situations, the engine's braking power, torque, and brake-specific fuel consumption were comparable to those of petro -diesel fuel.

Amin and Alireza, (2016), investigated how vegetables affected a diesel generator with a 1530 r/min engine speed. The study's findings indicated an increase in brake power, torque, and thermal efficiency as well as a decreasing tendency.

In a study by Sathiyamoorthi and Sankaranarayanan, (2017), in the VOD blend LGO25, 2.5 and 5% concentrations of ethanol were employed, with a 1:3 volume ratio of lemongrass oil to diesel. When ethanol is used instead of diesel or an LGO25 blend, the BSFC, BTE, NOx emission, heat release rate, and combustion pressure are all higher. With the usage of ethanol, smoke and HC emissions were shown to be reduced.

In a study of Qi *et al*. (2017), in the common rail direct injection CI engine, a tung oil-diesel-ethanol blend in various volume proportions generated by micro-emulsification was tested and studied. They discovered that when the quantity of ethanol in the blended fuels increases, the viscosity and density properties of the mixed fuels closely resemble those of

diesel. They've also admitted that ethanol has a significant impact on engine performance.

Maintaining the amount of diesel as 30% by volume, Prakash *et al*. (2018), 1:6:3, 2:5:3, and 3:4:3 blends of castor oil, ethanol, and diesel were used. The BTE, emission, and combustion properties of the ternary blend with 30% ethanol were comparable to those of diesel.

Kandasamy *et al*. (2019), examined the durability, performance, and emissions of a single cylinder high speed vehicle diesel engine running on ethanol blended cotton seed biodiesel, conducted an experimental examination. The durability test demonstrates that blending 20% ethanol to the base fuel has a negative impact on engine wear. Smoke and CO emissions have decreased as a result of the injection of ethanol, although unburned hydrocarbon emissions have increased. By volume, the base fuel is 5 percent biodiesel and 95 percent diesel.

A recent research project by Aguado-Deblas*et al*. (2020), and Dabi *et al*. (2021), diethyl ether is recommended as a VOD mix improver.

Dey *et al*. (2020), found that for diesel, palm biodiesel, and ethanol blends, the largest reduction in BSFC, NOx, and HC emissions occurs at 20% of engine load and for the D85BD10E5 (85, 10 and 5% diesel, palm biodiesel, and ethanol, respectively) blend.

Wu *et al*. (2020), with an increase in the amount of ethanol in the tetrahydrofuran (THF)-ethanol blend, the peak rate of heat release and BTE increased. However, under low loads, HC and CO emissions were

shown to increase. Ethanol was employed in the above-mentioned literatures in concentrations ranging from 2.5 to 40% by volume. However, VO generated from various feedstocks had varying viscosity and density levels, which had an impact on engine performance. As a result, the addition of ethanol to VOs derived from various feedstock is expected to affect engine behaviour in a different way.

A recent study by Vergel-Ortega *et al.* (2021), it was discovered that adding ethanol to diesel-biodiesel mixtures enhances thermal efficiency. With the addition of 2 and 4 percent, respectively, to the diesel biodiesel blend, the thermal efficiency increases by 0.8 and 1.2 percent. Exhaust gas analysis showed a decrease in HC, NOx, CO2, CO, and smoke opacity emissions.

In processing cassava tuber, the biomass left behind include peelings, leaves and roots, and in most cases, the biomass waste is not used for any economical purposes. Apart from the stem that may be used for re-planting, the peelings are usually disposed of indiscriminately, thus leading to environmental pollution. In the best cases, the current utilization of cassava peelings includes its potential re-placement for maize in animal feed, especially for ruminant animals. However, the drawback in this include; its low protein content (<6%), and the presence of anti-nutrients such as hydrocyanic acid, tannic acid, oxalic acid which could produce hazardous effects on ruminant animals. Consumption of cassava wastes is therefore greatly reduced, leaving the residue as non-valuable wastes in the environment compartment. However, the wastes

could be of tremendous value if utilized for energy generation.

In Nigeria, the quantity of cassava peelings generated is quite huge in places where garri (dried cassava flour), fufu (wet cassava flour) and Starch are produced and converting such wastes to energy resources would consume the wastes, sanitize the environment and benefit the nation economically. One of the ways to unlock this great potential of the waste is by getting valuable products out of it; hence, a reactor was being designed to hydrolyse the waste. In this research work, a percolation reactor was designed as described by to check its efficiency for the hydrolysis of agricultural waste such as cassava peelings.

Cassava peels ethanol yields more ethanol per unit of land area than other ethanol sources, making it a preferred biodiesel feedstock. Consequently, a lot of research has been done on the trans-esterification of cassava peels for the generation of biogas and biodiesel, and its ester in diesel engines has been studied in terms of engine performance and combustion properties (Yoon *et al.*, 2008; Ileri*et al.*, 2013; Cenk *et al.*, 2012). Roy *et al.*, (2013) tested biodiesel blends with CPE at high idle operations. In comparison to pure diesel fuel, the results demonstrated a great reduction in CO and HC emissions.

3.0 MATERIALS AND METHODS

The following materials and equipment are used in this research project work.

Materials

i. Diesel fuel

ii. Cassava peel ethanol

Equipment

iii. Engine test rig

iv. A measuring cylinders

v. Thermometers

vi. Dynamometer

vii. An analogue tachometer

viii. Fuel tank

3.1 Experimental Procedure

3.1.1 Preparation of bio-diesel blends using diesel and cassava peel ethanol

Cassava peel ethanol biodiesel is created by the trans-esterification method. The trans-esterification technique is used to produce bio-diesel from triglycerides. Food-grade vegetable oils, non-food-grade vegetable oils, animal fats, and waste restaurant greases can all be used to make

bio-diesel. This procedure is used to enhance the CPE ideal fuel qualities, such as density, kinematic viscosity, flash point, fire point, pour point, and calorific value. Trans-esterification, also known as alcoholysis, is the general word for the major family of chemical reactions in which an ester is changed into another ester through the interchange of alkyl groups. Trans-esterification is an equilibrium reaction that takes place when the reactants are mixed together. The presence of a catalyst, on the other hand, hastens the correction of the equilibrium. There are four basic phases in the trans-esterification process. Acid treatment followed by settling process, Base treatment followed by settling process, water washing and dehydration.

For this current experimental study three types of blends were prepared which were named as per there bio-fuel concentration. The blends are as following:

BLEND 1:– This blend contains 10% of cassava peel ethanol and 90% of diesel (B10).

BLEND 2:– This blend contains 20% of cassava peel ethanol and 80% of diesel (B20).

BLEND 3: – This blend contains 30% of cassava peel ethanol and 70% of diesel

(B30).

To ensure effective research analysis with the engine test rig, the bio-fuels were combined in the right percentage volume proportions. In this

study, the engine is driven with diesel fuel, ethanol, B10, B20, and B30

3.1.2 Determination of characteristic properties of the fuels used

The density, viscosity, cloud point, cetane number, flash point, boiling point, pour point, calorific value, moisture content, ash content, volatile matter content, fixed carbon, carbon content, hydrogen content, sulphur content, nitrogen content, oxygen content of the fuel samples were determined. The fuels analysis were classified as ultimate, proximate, and physical properties. Some of the equipment used for the fuel analysis include a viscometer, used to determine the viscosity of the fuels and a weighing balance, used for relative density of fuels.

3.1.3 Moisture or water content of fuels

The amount of water per unit mass of the dry solid is how a solid's moisture content is expressed. The ratio of the total mass of moisture in the original sample to the original wet mass of the sample also known as the wet basis water content is employed in this work. The presence of water can be detected in various ways. The traditional method of determining moisture is to dry a sample to a constant weight in an oven with air at 105 °C. Long-term drying in the oven might cause the biomass to decompose, distil, or oxidize, losing volatile components. Initial values are typically between 50% and 10%. During handling, the moisture content can fluctuate. The dried material was securely preserved in tight containers. Even quick handling must be done in an environment that is 100 percent dry. For accurate analytical results to be obtained, proper handling is necessary.

3.1.4 Proximate analysis

The fuel is divided into four categories by the proximate analysis: moisture (M), volatile matter (V), fixed carbon (FC), and ash. Proximate analysis consists of determination of ash content, moisture content, volatile matter, and fixed carbon by measuring the rate of weight loss of the sample as a function of time and temperature, the thermal behaviour of biomass has been examined (Thermo Gravimetric Analysis, TGA). The observed rates depend on the size and density of the sample in addition to time and temperature. The burning of the fixed carbon produces the ash content. Due to the oxidation procedure used to determine it, the resulting ash fraction is not an accurate representation of the original ash.

3.1.5 Ultimate analysis

The ultimate analysis typically provides information on the composition of elemental carbon (C), hydrogen (H), nitrogen (N), Sulphur (S), and oxygen (O) by comparing the differences in the fuel. An ultimate analysis is carried out and reported on a dry basis in order to minimize confusion and provide a good picture of the fuel itself, since otherwise moisture is shown as additional hydrogen and oxygen. To break down the sample into nitrogen, water, carbon dioxide, and sulphur dioxide, which are subsequently quantified by chromatography utilizing flame ionization or thermal conductivity detectors, the majority of the systems use catalytic combustion with pure oxygen. By catalytically converting oxygen to carbon monoxide, oxygen is determined.

3.1.6 Performance analysis using an RF4F 2.0l diesel engine turbo-charge

Diesel fuel (D100B0), blends of diesel and ethanol bio-diesel (B10 to B30) were tested for performance analysis using the RF4F diesel engine. The engine setup includes a fuel tank, dynamometer, two thermometers, a thermocouple, an analogue tachometer, dynamometer, and a battery. A dynamometer is coupled to the engine for loading. The cooling system has two channels for the inlet and outlet flow, to help maintain a good working temperature for the overall effective performance of the engine. An analogue tachometer was employed to measure the engine speed (in rpm). A measuring cylinder was employed to obtain the accurate volume of 100% diesel, biodiesel blends, and CPA required fuelling the engine. The experimental setup is shown in Figure 3.1.

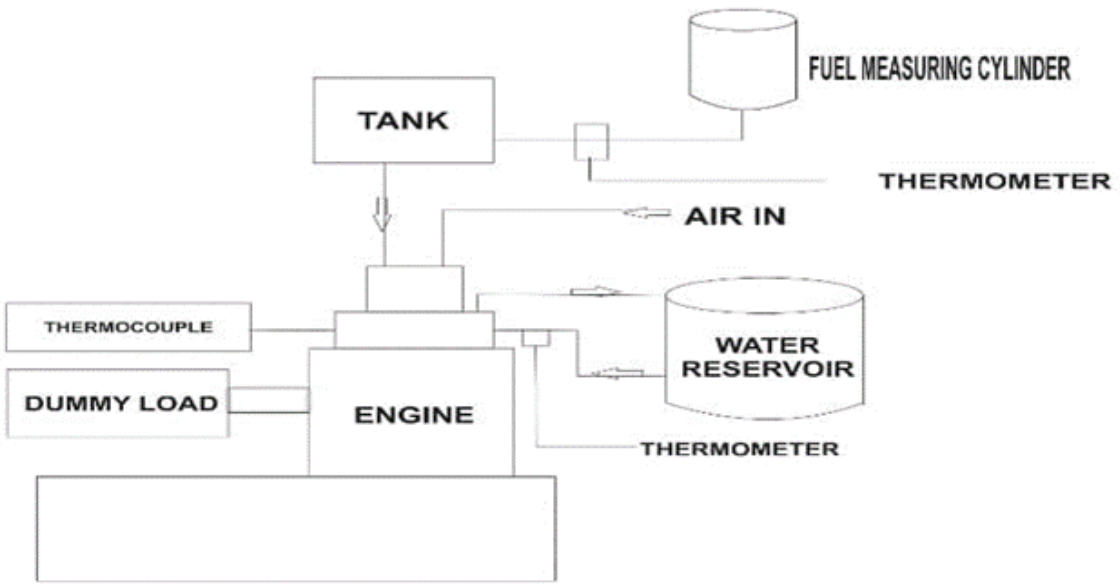


Figure 3.1: Experimental Setup

3.1.7 Performance test on the engine test rig

The following experiments were carried out on the RF4F turbocharged diesel engine.

i. Performance tests were conducted using diesel as fuel.

ii. Performance tests were conducted using 3 different blends of cassava peel ethanol (CPA) and diesel as fuel.

Diesel is blended with CPA in the following proportions by volume; 900 ml to 100 ml, 800 ml to 200 ml, 700 ml to 300 ml. This was effectively performed using a measuring cylinder. The blends were collected in different reservoirs and labelled according to its proportion by volume. The volume of fuel consumed was also determined, using a measuring cylinder.

The brake torque which is a major experimental factor in defining other parameters was first determined. A specific engine speed was set, and the engine was loaded gradually until it stopped. The load that stopped the engine alongside the radius gave the brake torque. This was done running the engine with 100% diesel, cassava ethanol, and the five blends. The time taken was determined using a stopwatch, and the fuel consumed was recorded. The readings were taken five times while using a particular fuel under increasing loads, and the average was calculated and recorded. The thermocouple was used to determine the exhaust gas temperature in °C.The two

thermometers were used to measure the temperature of the cooling water at inlet, the cooling water at outlet, and the ambient temperature. The engine load was gradually increased by adding weights while the engine speed was measured with an analogue, tachometer.

3.2 Engine Test Rig Specification

Table 3.1: A brief specification of the test engine

S/No	Parameter	Specification
1	Engine type	Diesel
2	Fuel preparation	Direct injection (DI)
3	Number of cylinder	4
4	Oil capacity	4.7 litres
5	Number of valves	16
6	Length of stroke	86 mm
7	Compression ratio	18.8:1
8	Engine displacement	2.0 litres
9	Power	100-110 HP (horse power) on 4000 rpm
10	Torque	220-230 Nm

3.3 Calculations to Determine the Parameters Required

3.3.1 Calculations to determine the indicated power

The entire power created within the cylinder throughout one whole cycle, ignoring any losses, is the indicated power of an internal combustion engine. It is the total amount of an engine's friction and braking power. Indicated Power is the power developed by the combustion of fuel in the cylinder of an engine. It is always more than brake power. It is, in essence, the amount of power that can be extracted from the expanding gases produced by the burning of fuel inside the cylinder while ignoring frictional losses, mechanical losses, and losses caused by heat and enthalpy.

The Indicated Power (IP) of an engine is determined mathematically by the formula;

$$IP = \frac{P_{mep}LANK}{60} \qquad 3.1$$

Where P is the mean effective gas pressure, N/m^2

P_{mep}, is the Mean effective pressure

L, is the length of stroke, m

A, is the area of piston, m^2

N, is the rotational speed of the engine, rpm

K, is the number of cylinders

Indicated power unit is Watts.

3.3.2 Calculations to determine the brake power

Brake power is the engine's useable power output, excluding the power needed to heat, fuel, or lubricate the engine or to circulate coolant to the engine. Brake power is the power available at the crankshaft of an engine.

The Brake Power (BP) of an engine is determined mathematically by the formula;

$$BP = \frac{2\pi NT}{60} \qquad 3.2$$

Where, π is taken to be 3.142

N, is the rotational speed of the engine

T, is Torque

Brake power is measured in Watts.

3.3.3 Calculations to determine the brake torque

Torque is the product of force and the radius arm. The ability of a force to alter a component's rotational motion is known as torque. It is the force applied to an object, and this force usually results in the object's rotational speed changing. It is the turning effect of a force about a point.

The Torque (T) of an engine is determined mathematically by the formula;

$$T = F \times r \qquad 3.3$$

Where T is Torque,

F, is Force

R, is radius

3.3.4 Calculations to determine the mechanical efficiency

Mechanical efficiency is the ratio of the brake power to the indicated power. It is the measure of the effectiveness with which a mechanical system operates. It is primarily the ratio of the power delivered by a mechanical system to the power supplied to it. The mechanical efficiency of an engine is always less than 1, due to friction.

The Mechanical Efficiency (M.E) of an engine is determined mathematically by the formula;

$$M.E = \frac{BP}{IP} \qquad 3.4$$

Where BP, is brake power,

IP, is indicated power

3.3.5 Calculations to determine the brake-specific fuel consumption

The brake-specific fuel consumption is a measure of the fuel efficiency

of an internal combustion engine that burns fuel to produces rotational power or shaft power.

Specific fuel consumption compares the efficiency of the internal combustion engine with the shaft output.

The brake-specific fuel consumption (Bsfc) of an engine is determined mathematically by the formula;

$$Bsfc = \frac{Mf}{BP} \qquad 3.5$$

where Mf is the mass flow rate.

Indicated power, brake power, brake torque, brake thermal efficiency, indicated thermal efficiency, mechanical efficiency, volumetric efficiency, and specific fuel consumption is calculated using the formulae stated above under each respective parameter.

3.3.6 Calculations

1. *Torque (Nm)*

a) Torque produced with diesel

Torque = Force (N) x Radius arm (m)

= 788.73×0.065

Torque = 51.3 Nm

b) Torque produced with B10

Torque = Force (N) x Radius arm (m)

= 656.29×0.065

Torque = 42.66Nm

c) Torque produced with B20

Torque = Force (N) x Radius arm (m)

= 715.15×0.065

Torque = 46.49Nm

d) Torque produced with B30

Torque = Force (N) x Radius arm (m)

=852.49×0.065

Torque = 55.41Nm

2. Brake power (kW), BP

a. The Brake power of diesel

$$Brake\ Power = \frac{2\pi NT}{60}$$

$$= \frac{2 \times 3.142 \times 400 \times 51.3}{60}$$

Brake Power = 2.149kW

b. The Brake power of B10

$$Brake\ Power = \frac{2\pi NT}{60}$$

$$= \frac{2 \times 3.142 \times 400 \times 42.66}{60}$$

$$Brake\ Power = 1.787kW$$

c. The Brake power of B20

$$Brake\ Power = \frac{2\pi NT}{60}$$

$$= \frac{2 \times 3.142 \times 400 \times 46.49}{60}$$

$$Brake\ Power = 1.948kW$$

d. The Brake power of B30

$$Brake\ Power = \frac{2\pi NT}{60}$$

$$= \frac{2 \times 3.142 \times 400 \times 55.41}{60}$$

$$Brake\ Power = 2.321kW$$

2. *The Mass flow rate of fuel (kg/hr),* Mf

$$Mf = \frac{\text{Mass of fuel (kg)}}{\text{Time (hr)}}$$

$$Mass = \text{Volume} \times \text{Density}$$

a) The mass flow rate of diesel (kg/hr);

$$Mass = 0.00008 \times 984.8$$

$$Mass = 0.07878\text{kg}$$

$$\text{Mf} = \frac{0.07878}{0.00833}$$

$$\text{Mf} = 9.457\text{kg/hr}$$

b) The mass flow rate of B10 (kg/hr);

$$Mass = 0.00008 \times 1014.2$$

$$Mass = 0.08114\text{kg}$$

$$\text{Mf} = \frac{0.08114}{0.0055}$$

$$\text{Mf} = 14.753\text{kg/hr}$$

The mass flow rate of B20 (kg/hr);

$$Mass = 0.000074 \times 1015.2$$

$$Mass = 0.07512\text{kg}$$

$$\text{Mf} = \frac{0.07512}{0.0055}$$

Mf = 13.658kg/hr

c) The mass flow rate of B30 (kg/hr);

Mass = 0.00008×927.6

Mass = 0.07421kg

$$Mf = \frac{0.07421}{0.0078}$$

Mf = 9.514kg/hr

3. Brake-specific fuel consumption

a. Brake-specific fuel consumption for diesel (kg/kWhr), bsfc

$$Bsfc = \frac{M_f}{BP}$$

$$= \frac{9.457}{2.149}$$

Bsfc = 4.4kg/kWhr

b. *Brake-specific fuel consumption for B10 (kg/kWhr), bsfc*

$$Bsfc = \frac{M_f}{BP}$$

$$= \frac{14.753}{1.787}$$

Bsfc = 8.256kg/kWhr

c. Brake-specific fuel consumption for B20 (kg/kWhr), bsfc

$$Bsfc = \frac{M_f}{BP}$$

$$= \frac{13.658}{1.948}$$

$$Bsfc = 7.011 kg/kWhr$$

d. Brake-specific fuel consumption for B30 (kg/kWhr), bsfc

$$Bsfc = \frac{M_f}{BP}$$

$$= \frac{9.514}{2.321}$$

$$Bsfc = 4.099 kg/kWhr$$

4. **Indicated power (kW),** $IP = \frac{P_{mep} LANK}{60}$

$$P_{mep} = \frac{2\pi n_r T}{n_c V_d} \quad (N/m^2)$$

Where n_r, is the number of crankshaft rotations for a complete engine cycle (for 4-stroke engine, $n_r = 2$)

n_c, is the number of cylinders ($n_c = 4$)

V_d, is the cylinder displacement ($V_d = 0.001998 m^3$)

T, s the Torque

i. P_{mep} (mean effective pressure) for diesel, (N/m^2)

$$P_{mep} = \frac{2\pi n_r T}{n_c V_d}$$

$$= \frac{2\times 3.142\times 2\times 51.3}{4\times 0.001998}$$

$$= \frac{644.74}{0.00799}$$

$$= 80.69 kN/m^2$$

ii. P_{mep} (mean effective pressure) for B10, (N/m^2)

$$P_{mep} = \frac{2\pi n_r T}{n_c V_d}$$

$$= \frac{2\times 3.142\times 2\times 42.66}{4\times 0.001998}$$

$$= \frac{536.15}{0.00799}$$

$$= 67.1 kN/m^2$$

iii. P_{mep} (mean effective pressure) for B20, (N/m^2)

$$P_{mep} = \frac{2\pi n_r T}{n_c V_d}$$

$$= \frac{2\times3.142\times2\times46.49}{4\times0.001998}$$

$$= \frac{584.3}{0.00799}$$

$$= 73.13kN/m^2$$

iv. P_{mep} (mean effective pressure) for B30, (N/m^2)

$$P_{mep} = \frac{2\pi n_r T}{n_c V_d}$$

$$= \frac{2\times3.142\times2\times55.41}{4\times0.001998}$$

$$= \frac{696.39}{0.00799}$$

$$= 87.16kN/m^2$$

N/2= 400/2= 200rpm

L= length of stroke (86mm= 0.086m)

A= area of cylinder $\frac{\pi D^2}{4}$, where (D= 0.086m) = $0.0058m^2$

K= 4 cylinders

a. Indicated power for diesel (kW), IP

$$IP = \frac{PLANK}{60}$$

$$IP = 3.3kW$$

b. Indicated power for B10 (kW), IP

$$IP = \frac{PLANK}{60}$$

$$IP = 2.938kW$$

c. Indicated power for B20(kW), IP

$$IP = \frac{PLANK}{60}$$

$$IP = 3.099kW$$

d. Indicated power for B30 (kW), IP

$$IP = \frac{PLANK}{60}$$

$$IP = 3.48kW$$

7. Mechanical efficiency, $M.E = \frac{BP}{IP}$

a. Mechanical efficiency for diesel

$$M.E = \frac{BP}{IP}$$

$$= \frac{2.149}{3.3} \times 100$$

$$M.\ E = 65.12\%$$

b. Mechanical efficiency for B10

$$M.\ E = \frac{BP}{IP} \times 100$$

$$= \frac{1.787}{2.938} \times 100$$

$$M.\ E = 60.82\%$$

c. Mechanical efficiency for B20

$$M.\ E = \frac{BP}{IP} \times 100$$

$$= \frac{1.948}{3.099} \times 100$$

$$M.\ E = 62.86\%$$

d. Mechanical efficiency for B30

$$M.\ E = \frac{BP}{IP} \times 100$$

$$= \frac{2.321}{3.48} \times 100$$

$$M.\ E = 66.71\%$$

3.4 Bill of Engineering Measurement and Evaluation (BEME)

The cost required to perform this research project work is very important so as to determine its effectiveness, accuracy, and results. The billing includes the cost of materials, equipment, test analysis, and cost of labour. The table below shows the Bill of Engineering Measurement and Evaluation (BEME)

Table 3.2: Bill of Engineering Measurement and Evaluation (BEME)

S/N	Material	Quantity	Unit	Rate (N)	Amount (N)
1	DIESEL	3.7	LITRE	1200	4600
2	CASSAVA PEEL	2	KG	5000	10000
3	ESTERIFICATION OIL (CATALYST)	2	CL	1000	2000
4	RUBBER HOSE	1.5	YARD	2000	3000
5	WATER	5	GALLON	100	500
6	FUEL	6.6	LITRE	750	5000
7	TRANSPORTATION			500	1000
8	TOTAL				**25,100**

4.0 RESULTS AND DISCUSSION OF RESULT

4.1 Results

4.1.1 Blends

Diesel was blended with CPE in the following proportions by volume;

900 ml to 100 ml, 80 0ml to 200 ml, 700 ml to 300 ml. This was effectively performed using a measuring cylinder. The measuring cylinder has adequate and accurate graduations which enabled the right amount of fuel to be measured to prepare the blends. The blends were collected in different reservoirs and labelled according to its proportion by volume (B10, B20, and B30).

The fuels (diesel, CPE biodiesel blends) were used to run the engine. The brake torque, brake power, brake thermal efficiency, indicated power, indicated thermal efficiency, mechanical efficiency were determined for each fuel.

4.1.2 Calculated values from test

The performance readings were obtained from the test, and is shown in Appendix

4.2 Discussion of Results

4.2.1 Proximate analysis of the fuels

Table 4.1 shows the proximate analysis of the fuels, with B20 having the highest moisture content. 100% diesel has moisture content a bit lower than B20. B30 blend has moisture content a bit lower than diesel. The moisture content of diesel is a bit higher than that of B30. B20 has the lowest ash content. B20 has the lowest volatile matter content.

Table 4.1: Proximate analysis of the diesel, and the blends

Properties	Diesel	B10	B20	B30
Moisture content	14.25	13.12	19.08	14.10
Ash content	2.0	1.05	1.00	1.55
Volatile matter content	38.12	32.17	24.33	32.14
Fixed carbon content	46.13	54.66	56.54	53.26

4.2.2 Ultimate Analysis of the Fuels

Table 4.2 shows the ultimate analysis of the fuels. The B20 blend has the lowest carbon content. The fossil content on ethanol is higher than the other fuels. The B20 blend composition of hydrogen is the lowest. Diesel, B10, and B30 all has no Sulphur composition. The liebig condenser is attached to the furnace to determine Sulphur, and oxygen content of the fuels. The kjeldahl distillation instrument was used for distillation to purify the fuels.

Table 4.2: Ultimate analysis of the diesel, and the blends

Properties	Diesel	B10	B20	B30
Carbon	39.8	42.12	39.16	41.72
Hydrogen	5.4	4.2	4.0	5.1
Sulphur	0.0	0.0	0.5	0.0
Nitrogen	1.0	1.0	1.0	0.5
Oxygen	40.0	38.96	41.02	37.57

4.2.3 Physical properties of fuel

Fuel is the source of energy that burns in air with sufficient amount of oxygen. The fuel can be in solid, liquid or gaseous form. The properties of fuel are very important for determining the fuel efficiency as well as its exhaust emissions. It also helps in optimizing the consumption of energy during combustion. The understanding of the fuel properties will able to choose a right fuel for the use and thereby improving engine performance. The properties vary with the type of fuel. The liquid fuels have properties different from solid and gaseous fuels.

Table 4.3 shows the physical properties of the fuels. The diesel has a

viscosity that is a bit lower than the viscosity of blends. The B10 blend was the least viscous fuel having a viscosity of 9.0 cst. A viscometer was used to determine the viscosity of the fuels. The B30 blend has the least density of 0.9276 kg/mm^3. An electronic balance together with the volume of the fuel measured was used to determine the density of the fuels.

Table 4.3: Physical properties of the diesel, cassava peel ethanol and the blends

Properties	Diesel	B10	B20	B30
Viscosity (cSt)	39.0	9.0	16.0	18.0
Density (kg/mm^3)	0.9848	1.0142	1.0152	0.9276
Flashpoint (°C)	110	140	156	178
Boiling point (°C)	68.0	120	148	159
Calorific value (kJ/kg)	44,000	45,500	44,000	44,500

4.2.4 Calculated values from the engine performance test

The engine was set to 600 rpm. The highest torque (55.41 Nm) was produced when B30 was used as fuel in the RF4F turbocharger diesel engine, the torque was greater than that produced when diesel was used as fuel in the engine with B10 having the lowest torque (42.66 Nm).

The B30 blend produced the highest brake power (2.321 kW) in the RF4F turbo charger diesel engine, the brake power was greater than that produced when diesel was used as fuel in the engine with B10 having the lowest brake power (1.787 kW). The mass flow rate was highest (14.753 kg/hr) when B10 was used as fuel in the engine, with diesel having the lowest mass flow rate of 9.457kg/hr. The indicated power was highest (3.48kW) when B30 was used as fuel in the engine, with B10 having the lowest indicated power of 2.938kW. The Bsfc was highest (8.256 kg/kWhr) when B10 was used as fuel in the engine, with B30 having the lowest Bsfc of 4.099 kg/kWhr. The mean effective pressure was highest (87.16 kN/m^2) when B30 was used as fuel in the engine, with B10 having the lowest mean effective pressure of 67.1kN/m^2. The brake thermal efficiency was highest when B30 was used as fuel in the engine, with B10 having the lowest brake thermal efficiency. The indicated thermal efficiency was highest when B30 was used as fuel in the engine, which was greater than that of diesel; with B10 having the lowest indicated thermal efficiency. The mechanical efficiency (66.71%) was highest when B30 was used as fuel in the engine, which was greater than that of diesel, with B10 having the lowest mechanical efficiency of 60.82%

Table 4.4: Calculated Values from the Engine Performance Test

Fuels	Torque (Nm)	Brake power (kW)	Mf (kg/h)	IP (kW)	bsfc (kg/kWh)	P_{mep} (kN/m^2)	M.E (%)	Load (N)

Diesel	51.3	2.149	9.457	3.3	4.4	80.69	65.12	946.67
B10	42.66	1.787	14.753	2.938	8.256	67.1	60.82	814.23
B20	46.49	1.948	13.658	3.099	7.011	73.13	62.86	873.09
B30	55.41	2.321	9.514	3.48	4.099	87.16	66.71	1,010.43

4.3 Comparative Analysis Results

4.3.1 Brake power

The 100% diesel fuel was tested first, followed by the biodiesel blends B10, B20, and B30, with engine loads ranging from 519.93N to 946.67N. In this study, it was found out that brake power improves as load increases. B30 had the best brake power amongst all the fuels, and next to B30 was the 100% diesel. At the set engine speed of 600rpm B30 had the maximum brake power of 2.321kW, greater than the brake power of diesel which was 2.149kW. This study shows that as load increases brake power increases and as load decreases brake power decreases.

4.3.2 Engine torque

By increasing engine loads, cassava peel ethanol bio-diesel B10, B20, and B30 blends, and CPE were tested after pure diesel fuel. It was discovered that B30 had a greater torque higher than diesel, and the other fuels. At the set engine speed of 600rpm B30 had the maximum

torque of 55.41Nm, greater than the torque of diesel which was 51.3Nm. The biodiesel B30 had the best torque of all the fuels, while the 100% diesel was right behind B30.

4.3.3 Indicated power

The indicated power began to increase as the engine became loaded gradually. In this research project work, it was found that for all operations involving diesel, cassava peel ethanol, and the biodiesel blends, the indicated power increased as the load increased. This increase might be brought on by the fact that the mean effective gas pressure increased with increasing load. There was an observed rise in the mean effective pressure as the load increased, causing the increase in indicated power. B30 had the highest mean effective pressure, resulting in the highest indicated power developed in the engine. B30 had the highest indicated power of 3.48 kW, greater than the indicated power of diesel which was 3.3 kW.

4.3.4 Brake-specific fuel consumption

Brake specific fuel consumption (Bsfc), determines how much fuel an engine will use per hour for each power it generates. It is a measure of the fuel flow rate per unit of power output. For compression ignition engines, the Bsfc can be as low as 0.2kg/kWh, which is obviously desired. A low Bsfc is more efficient, as high Bsfc are less efficient. Brake-specific fuel consumption changes with engine load and speed (rpm). The Bsfc began to drop sharply as the engine became loaded.

In this research project work, it was found that for all operations involving diesel, CPE, and bio-diesel blends, the brake-specific fuel consumption decreased as the load increased. This decrease might be brought on by the fact that fuel consumption increased more slowly than brake power with load. Additionally, as load increases, the temperature of the engine increased as well, reducing the ignition delay. By reducing the ignition delay, combustion is improved and fuel consumption is decreased. However, with lower loads compared to higher loads, the rate of decrease in brake-specific fuel consumption was greater. B10, B20 used more fuel compared to the 100% diesel while fully loaded to 946.67N respectively.

For a particular blend, the specific fuel consumption is seen to decline as the load rises, whereas B10 shows somewhat higher specific fuel consumption for the same load. The biodiesel B10's maximum brake specific fuel consumption was 8.256 kg/kWhr more than diesel which had a maximum brake specific fuel consumption of 4.4 kg/kWhr.

4.3.5 Mechanical efficiency

In this research project work, it was found that for all operations involving diesel, cassava peel ethanol, and the biodiesel blends, the mechanical efficiency increased as the load increased. This increase might be brought on by the fact that the brake power, and indicated power increased with increasing load. B40, and CPE had mechanical efficiencies of 64.55% at full load which was close to the mechanical efficiency of diesel. B30 had the highest mechanical efficiency of 66.71%,

greater than the mechanical efficiency of diesel which was 65.12%. At full load, B30 produced mechanical efficiency greater than that of all the fuels.

5.0 CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The following conclusion was drawn from the performance of diesel and its blend with cassava peel ethanol using an engine test rig, which was based on the observation, and evaluation of results that was made in the course of carrying out this project research work.

1. The fuel was prepared and was used to run the diesel engine. B30 blend gave the best overall performance, which was even higher than 100% diesel. Biodiesel B30 blend produced the greatest amount of power. This study led to the conclusion that cassava peel ethanol biodiesel is a promising replacement fuel for diesel engines. Based on the findings, it can be concluded that blend B30 operated efficiently

2. The engine was set to 600rpm, and the fuels were used to run the engine. The B30 blend gave the best overall performance, producing the highest torque, brake power, mean effective pressure of gas, indicated thermal efficiency, brake thermal efficiency, and mechanical efficiency. The brake-specific fuel consumption Bsfc which is best at the lowest was produced using B30. B30 mass flow rate was slightly higher than that of the 100% diesel fuel. The 100% diesel had higher brake-specific fuel consumption than B30 biodiesel. B30 was found to have the lowest brake-specific fuel consumption (Bsfc) across the board in terms of engine performance. According to the results, all of the blends had inferior engine performance in

terms of Bsfc than B30 blends.

3. The outcome shows that biodiesel B30 brake power, torque, indicated power, indicated thermal efficiency, brake thermal efficiency, mechanical and efficiency was greater than that of diesel. The brake power of diesel was only greater than the other fuels. B10 biodiesel had the least brake power. Based on the results of this study, it can be inferred from test results that all blends worked safely and efficiently within the engine test rig but B30 blend gave the optimum performance within the tested diesel engine. B30 performed with the highest efficiency

4. This study's findings noticed that biodiesel B10 had the lowest torque, brake power, mean effective pressure, indicated power, and indicated thermal efficiency, brake thermal efficiency, and mechanical efficiency of which all were lower than that of the diesel fuel.

5. The biodiesel B10, B20 produced torque, brake power, indicated power, mean effective pressure, brake thermal efficiency, indicated thermal efficiency and mechanical efficiency lower than that of the 100% diesel. Cassava peel ethanol biodiesel B30 mix produced the greatest amount of power. This study led to the conclusion that cassava peel ethanol biodiesel is a promising replacement fuel for diesel engines.

5.2 Recommendations

The following recommendations were made.

1. It is recommended that the B30 blend be used to run diesel engines, because it gives the best engine performance.

2. It is recommended that a high precision dynamometer be used to carry out the test on the diesel engine.

3. It is recommended that the institution gets the equipment used for the test analysis which includes the viscometer, weighing balance, oven instrument, and furnace.

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APPENDIX A

Table A: Shows the variation of temperature with respect to load (519.93N)

Temperature (℃)	Diesel	B10	B20	B30
Cooling water inlet (℃)	49.667	31.33	60.33	63
Cooling water outlet (℃)	57	46.667	82	89
Exhaust (℃)	128	126.33	114.67	106.67

APPENDIX B

Table B: Shows the variation of temperature with respect to load (711.225N)

Temperature (℃)	Diesel	B10	B20	B30
Cooling water inlet (℃)	60.33	40.667	66.33	70
Cooling water outlet (℃)	69.33	60.33	90	97.667
Exhaust (℃)	137.33	129.667	119.33	114.667

APPENDIX C

Table C: Shows the variation of temperature with respect to load (843.66N)

Temperature (℃)	Diesel	B10	B20	B30
Cooling water inlet (°C)	63.33	50.33	69	74.33
Cooling water outlet (°C)	79.667	66.667	94	104
Exhaust (°C)	140.3	138.67	138	139.667

APPENDIX D

Table D: Shows the variation of temperature with respect to load (946.67N)

Temperature (℃)	Diesel	B10	B20	B30
Cooling water inlet (°C)	78	58.667	76	76.667
Cooling water outlet (°C)	83	79	97.667	107.667
Exhaust (°C)	145	148.677	146.8	140.3



Trans-esterification experimental setup Image